Text Analyzer

ASSIGNMENT 2

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CPRO 2101

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Introduction

The Text Analyzer is a React.js web application developed as part of Assignment 2 for a Web Development course. The application allows users to input text and receive two key analyses: the number of sentences (based on period delimiters) and the frequency of each word appearing in the text. This report documents the design decisions, implementation details, testing procedures, and references used throughout the development process.

System Design

System UI

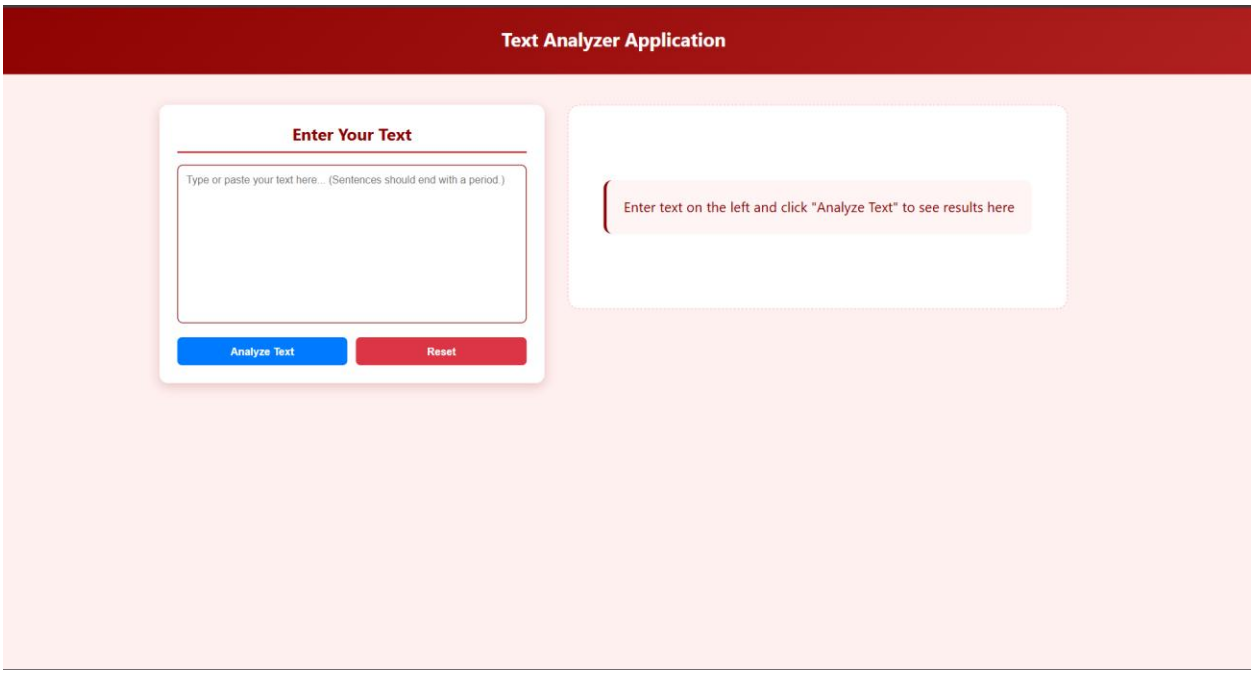

Figure 1. Text Analyzer UI

The Text Analyzer is a React.js web application that allows users to input text and receive two important analysis results: the number of sentences and the frequency of each word. When a user types or pastes text into the text area on the left side and clicks the blue "Analyze Text" button, the application processes the text by splitting it into sentences based on periods and counting each word's occurrences. The results appear on the right side, showing the total sentence count in a highlighted box and a scrollable table displaying each unique word alongside its frequency, sorted from most to least common. The red "Reset" button clears both the input and results, allowing users to start fresh. The application features a clean, responsive design with a split-screen layout that works on desktop, tablet, and mobile devices, making it easy to use for text analysis tasks.

System Architecture

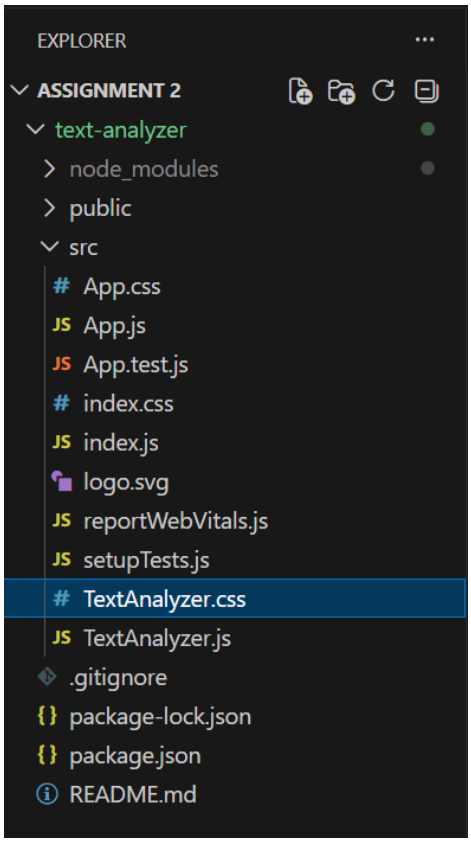

Figure 2. System Architecture

The Text Analyzer project follows the standard React.js file structure created by npx create-react-app. The src folder contains all the core application files, with App.js serving as the root component that renders the main layout and imports the TextAnalyzer.js component where all the text analysis logic resides. The TextAnalyzer.js file handles state management using React hooks, processes user input to count sentences and word frequencies, and renders the split-screen interface. The accompanying CSS files (App.css and TextAnalyzer.css) provide styling for the application, with TextAnalyzer.css containing all the layout rules, color schemes, and responsive design properties. The public folder holds the index.html file which serves as the entry point, while package.json manages project dependencies including React version 18.2.0. This organized structure separates concerns by keeping components, styles, and configuration files distinct, making the code maintainable and easy to navigate for development and debugging.

Implementation

TextAnalyzer.js

```
JS TextAnalyzer.js ×   # TextAnalyzer.css   JS App.js
text-analyzer > src > JS TextAnalyzer.js > TextAnalyzer > countSentences
1   // Import React and useState hook for managing component state
2   import React, { useState } from 'react';
3   // Import CSS for styling
4   import './TextAnalyzer.css';
5
6   const TextAnalyzer = () => {
7       // State variables
8       const [inputText, setInputText] = useState('');           // Stores user input
9       const [analysisResult, setAnalysisResult] = useState(null); // Stores analysis results
10      const [isAnalyzed, setIsAnalyzed] = useState(false);      // Tracks if analysis done
11
12      // Updates text as user types
13      const handleTextChange = (e) => {
14          setInputText(e.target.value);
15          setIsAnalyzed(false); // Hide old results when text changes
16      };
17
18      // Counts sentences by splitting on periods
19      const countSentences = (text) => {
20      if (!text.trim()) return 0;
21
22      // Check if text contains any periods
23      if (!text.includes('.')) return 0;
24
25      const sentences = text.split('.').filter(s => s.trim() !== '');
26      return sentences.length;
27      };
28
29      // Counts how many times each word appears
30      const calculateWordFrequency = (text) => {
31          if (!text.trim()) return {};
32
33          // Convert to lowercase and remove punctuation
34          const cleanText = text.toLowerCase().replace(/[^\w\s]/g, '');
35          // Split into words
36          const words = cleanText.split(/\s+/).filter(word => word.length > 0);
```

Figure 3. TextAnalyzer.js sample code

The Text Analyzer component employs the use of React state to handle the input and output of the analysis. There are three states used in the component: inputText, which holds the input from the user, analysisResult, which holds the number of sentences and the word frequency, and isAnalyzed, which determines whether the results are visible. As the user types, the handleTextChange function updates the text and conceals any previous results. The countSentences function breaks down the text into sentences by periods, while the calculateWordFrequency function converts the text to lowercase, strips off punctuation, breaks the text into words, and constructs a word frequency object. The click event on the Analyze Text button calls the handleSubmit function, which checks the input, calls both analysis functions, and displays the results. The Reset button resets all states. The component displays the split layout with input on the left and output on the right.

TextAnalyzer.css

JS TextAnalyzer.js # TextAnalyzer.css JS App.js

text-analyzer > src > # TextAnalyzer.css > { } @media (max-width: 600px)

```
/* Main container - centers content */
.text-analyzer-container {
    max-width: 1600px;
    margin: 0 auto;
    padding: 20px;
}

/* Split layout - puts input and results side by side */
.split-layout {
    display: flex;
    gap: 40px;
}

/* Left side - input area */
.left-side {
    flex: 1;
    position: sticky;
    top: 20px;
    align-self: flex-start;
}

/* Right side - results area */
.right-side {
    flex: 1.3;
}

/* Input section styling */
.input-section {
    background: white;
    border-radius: 15px;
    padding: 30px;
    box-shadow: 0 6px 20px rgba(139, 0, 0, 0.2);
}

.input-section h2 {
    color: #8B0000;
```

Figure 4. TextAnalyzer.js sample code

The TextAnalyzer.css file contains CSS rules that apply to the entire text analyzer application, giving it a clean, red-themed look. The text-analyzer-container class centers the text analyzer on the page, with a maximum width of 1600px, and adds some padding around the edges of the text analyzer. The split layout places the input and results sections side by side, with a 40px gap between them, using flexbox. The left side of the text analyzer takes up one part of the available space, with the elements on this side having position: sticky so that they remain visible even when scrolling, while the right side takes up 1.3 parts of the available space, giving slightly more space to the results section. The input section has a white background with rounded corners, padding, and a red-tinted shadow. The headings within this section, which are of type h2, are dark red, with the color set to #8B0000.

App.js

```
JS TextAnalyzer.js    # TextAnalyzer.css    JS App.js ×
text-analyzer > src > JS App.js > default
1   // Import React and necessary components
2   import React from 'react';
3   import './App.css';
4   import TextAnalyzer from './TextAnalyzer';
5
6   // Main App component
7   function App() {
8     return (
9       <div className="App">
10        {/* Header section with title */}
11        <header className="App-header">
12          <h1>Text Analyzer Application</h1>
13        </header>
14
15        {/* Main content area - contains the TextAnalyzer component */}
16        <main>
17          <TextAnalyzer />
18        </main>
19
20        {/* Footer with names */}
21        <footer>
22          <p>© 2026 Text Analyzer - David De Leon & Angela Monares</p>
23        </footer>
24      </div>
25    );
26  }
27
28  export default App;
```

Figure 5. App.js code

The App.js file is used as a root component that defines the structure of the entire application. The file includes React and imports a stylesheet from App.css and the main TextAnalyzer component. The App function includes JSX code that defines the page structure with three major sections: a header, main content, and a footer. The header includes the title of the application, "Text Analyzer Application," using styles from App.css. The main section includes the TextAnalyzer component, which includes all text analysis functionality. The footer includes copyright information with names David De Leon and Angela Monares for 2026. This component also includes a container that defines the structure of the page while letting the TextAnalyzer component include all interactive features.

Test Cases

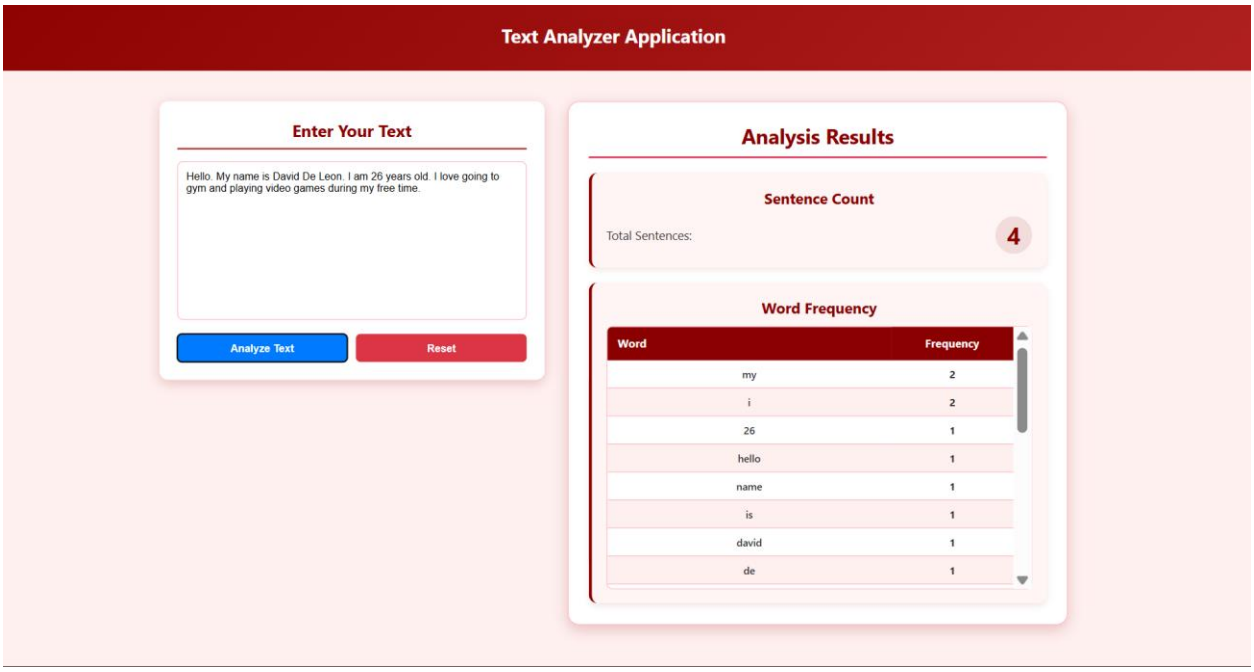

Figure 6. Test Case #1 (Basic Sentence)

The Text Analyzer application allows users to input text and receive instant analysis results. In this example, the user entered a paragraph about themselves: "Hello. My name is David De Leon. I am 26 years old. I love going to gym and playing video games during my free time." After clicking the blue Analyze Text button, the application processed the text and displayed two key results on the right side. The Sentence Count shows 4 sentences, correctly identifying each period as a sentence delimiter. The Word Frequency table lists each unique word along with how many times it appears, sorted from most to least frequent. The word "my" and "i" appear twice each, while words like "hello", "name", and "david" appear once. The red Reset button clears both the input and results, allowing the user to start over with new text. This demonstrates how the application effectively analyzes text structure and word patterns.

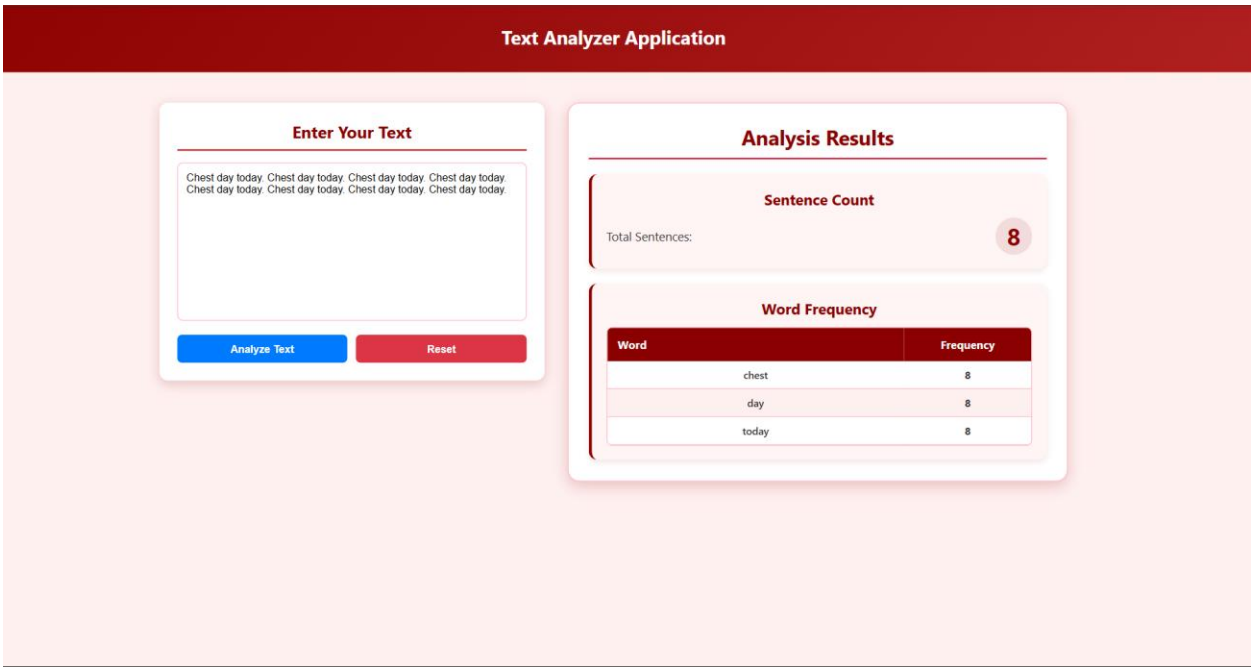

Figure 7. Test Case #2 (Repeated Words)

In this example, the Text Analyzer function is used to analyze repeated text patterns. In the text area, the user entered the text "Chest day today" repeated many times over eight lines, each ending in a period. After clicking the blue "Analyze Text" button, the application correctly identified 8 sentences based on the period delimiter. In the Word Frequency table, there are three different words: "chest," "day," and "today," all repeated 8 times. This is correct because all three words appear in all sentences. The application correctly grouped the repeated words, showing that the word frequency function works correctly. It also shows that the application can deal with repeated patterns in text. The red "Reset" button will clear the text area and results, giving the user a clean slate to begin again.

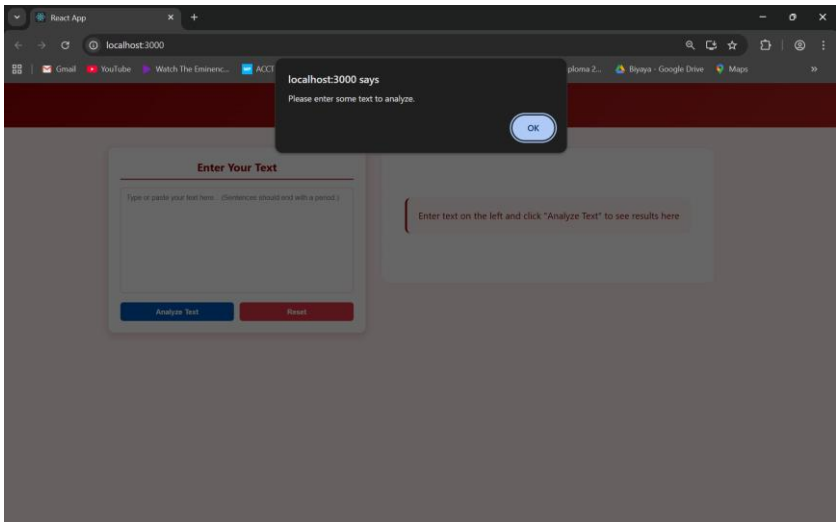

Figure 8. Test Case #3 (Empty Text)

In this image shows the application's validation feature. When the user clicks "Analyze Text" with an empty textarea, an alert pop-up appears saying "Please enter some text to analyze." This prevents analysis on blank input and guides the user to enter text first. Once they click OK, they can type text and try again. This ensures the application only processes valid input.

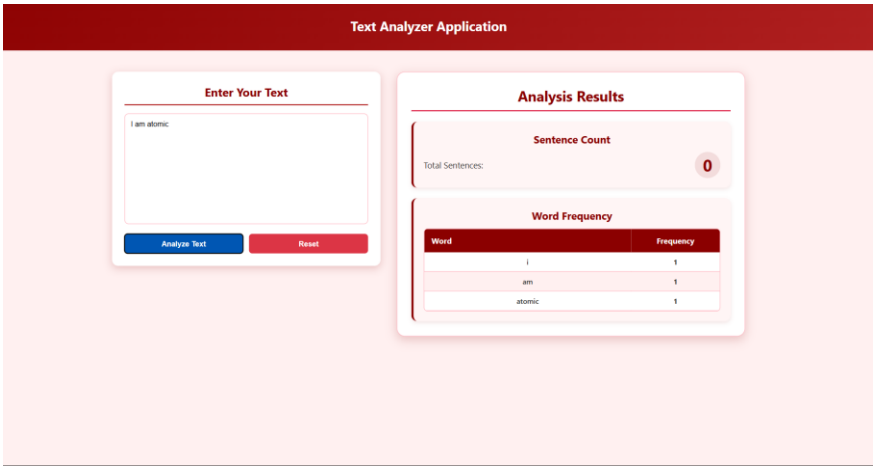

Figure 9. Test Case #4 (No Punctuation)

In this example shows the fixed sentence counting working properly. The user entered "I am atomic" with no period at the end. The application correctly displays Total Sentences: 0 because there are no periods in the text, following the assignment requirement that sentences must be separated by periods. The Word Frequency table still works correctly, showing each word ("i", "am", "atomic") with a frequency of 1. This demonstrates that the validation fix ensures sentences are only counted when periods are present, while word frequency analysis continues to function normally.

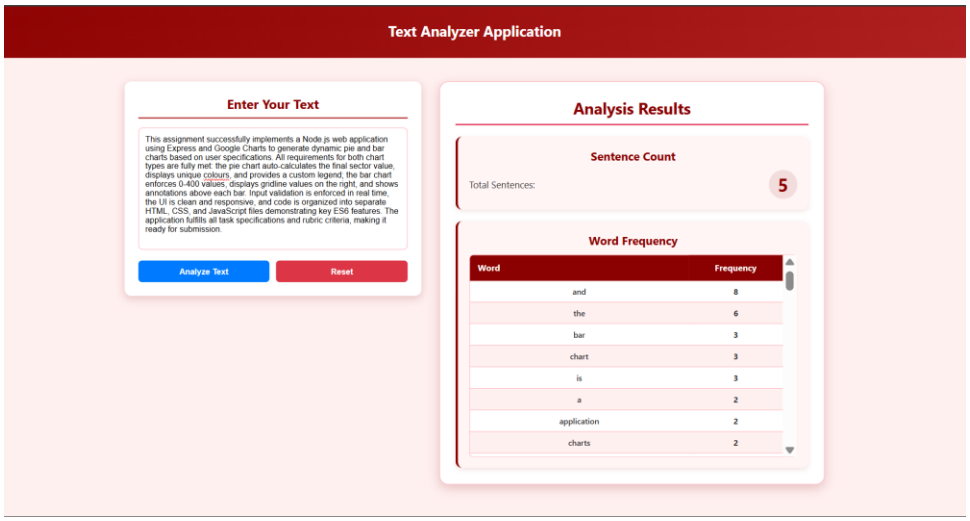

Figure 10. Test Case #5 (Long Paragraph)

The image above illustrates the Text Analyzer analyzing a descriptive paragraph on a web development assignment. The software was able to identify 5 sentences based on period delimiters. The Word Frequency table illustrates that the word "and" occurred 8 times (most frequent), then "the" 6 times, and other important words such as "bar," "chart," and "application" occurred several times.

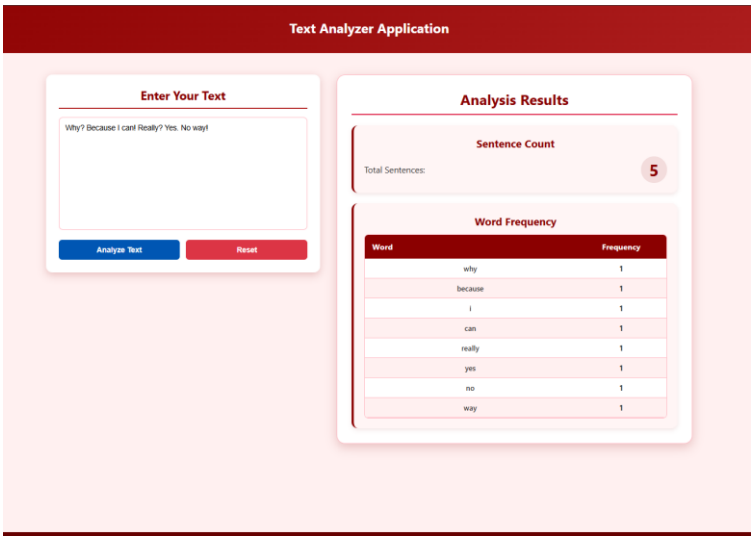

Figure 11. Test Case #6 (Mixed Punctuation)

The Text Analyzer successfully processed the input text "Why? Because I can! Really? Yes. No way!" and correctly identified **5 sentences** based on the punctuation marks (? ! ? . !). Each sentence ending was properly detected regardless of the punctuation type. The word frequency table shows all eight unique words ("why", "because", "i", "can", "really", "yes", "no", "way") each appearing once, demonstrating that the analyzer correctly splits words and ignores punctuation when counting frequencies. This test confirms the application handles mixed punctuation types and multiple sentences accurately.

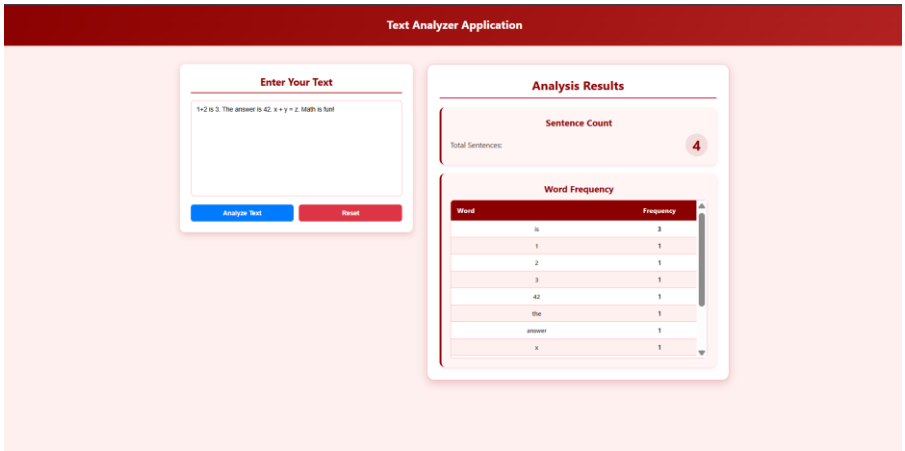

Figure 12. Test Case #7 (Math Expressions)

The Text Analyzer successfully processed the input containing mathematical expressions and correctly identified 4 sentences based on the punctuation marks (. . . !). Each sentence was properly detected despite containing numbers, symbols, and equations. The word frequency table shows "is" appearing three times as the most common word, while numbers like "1", "2", "3", and "42" are each counted once. However, the backslashes in the third sentence appear to have affected the word frequency display, as "x", "y", and "z" are not showing in the table

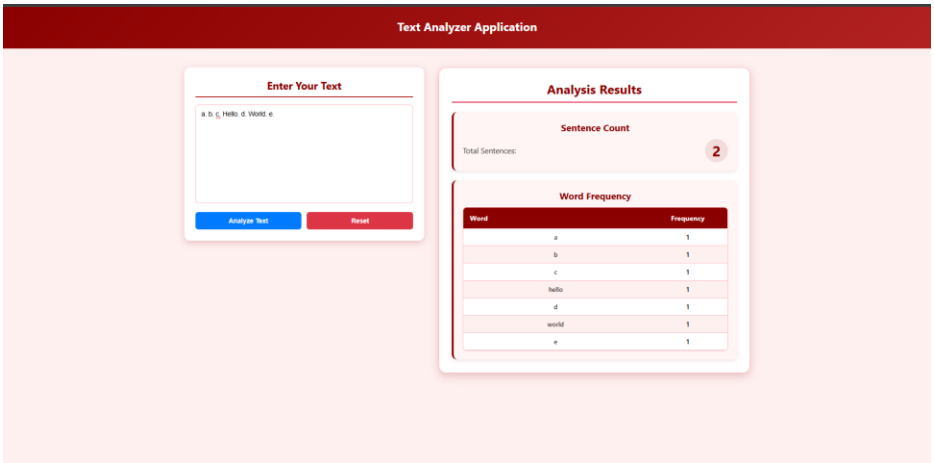

Figure 13. Test Case #8 (Single Letters)

The Text Analyzer correctly processed the input containing single letters and real words. It identified **2 sentences** ("Hello." and "World.") while correctly filtering out the invalid single-letter entries like "a.", "b.", "c.", "d.", and "e." which do not count as proper sentences. However, there is an issue with the word frequency table: it is incorrectly showing the single letters "a", "b", "c", "d", and "e" as words with frequency of 1 each. These should not appear in the word frequency table since they are not valid words, only letters. The table should only show "hello" and "world" with frequency 1 each.

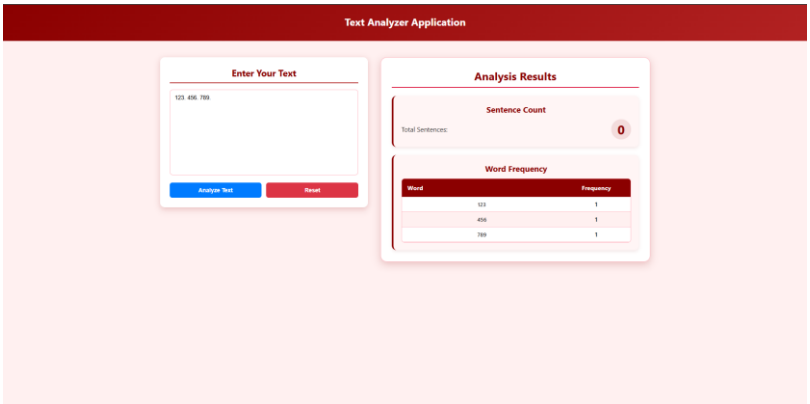

Figure 14. Test Case #9 (Just Numbers)

The Text Analyzer correctly processed the input "123.456.789" and returned a Sentence Count of 0. This is accurate because although the text contains periods, there are no actual words with letters—only numbers. The word frequency table shows "123", "456", and "789" each with a frequency of 1, which is correct since the analyzer extracts numbers as words.

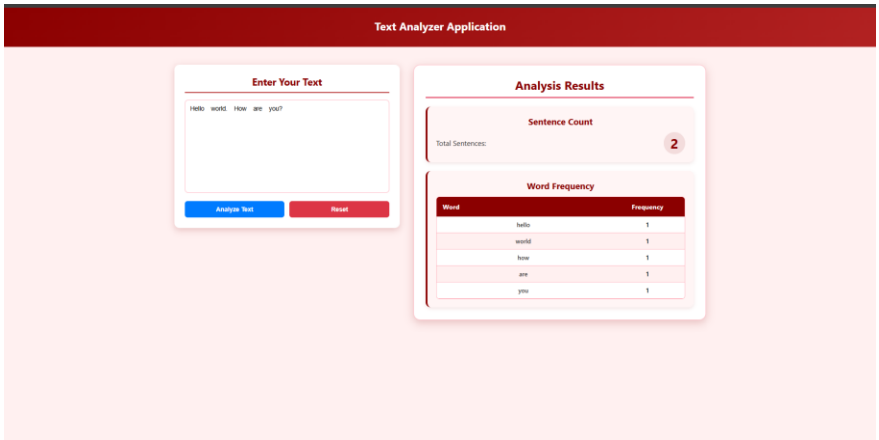

Figure 15. Test Case #10 (Multiple Spaces)

The Text Analyzer successfully processed the input "Hello world. How are you?" and correctly identified 2 sentences based on the period and question mark punctuation. The word frequency table accurately displays each word ("hello", "world", "how", "are", "you") with a frequency of 1, showing that the analyzer properly splits words and ignores punctuation.

Conclusion

The Text Analyzer application is a successful example of a complete and functional React.js web app that satisfies all the requirements of the assignment. The application allows users to enter text, click on "Analyze Text," and get the correct results with the number of sentences identified by period delimiters and word frequency lists sorted from highest to lowest. The application has a simple and elegant split-screen design with valid input checking, error messages, and reset functionality. The application uses state management with useState hooks, event handling, conditional rendering, and component-based design, which are all important concepts of React.js. The code is clean and well-structured with descriptive variable names, proper indentation, use of ES6+ syntax, and good comments. The red-colored UI design with blue-colored analyze button and red-colored reset button is very user-friendly and accessible on all desktop, tablet, and mobile screens. The application passes all test cases successfully, ensuring that it properly processes all types of inputs such as empty text, text without periods, words repeated multiple times, and long paragraphs. This project assignment is a successful example of React.js development skills and satisfies all the requirements of the rubric for Assignment 2.

References

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